Loan Approval Dataset Preprocessing and Modelling

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INTRODUCTION

- In high risk domains like finance, ensuring the reliability of loan approvals is important.
- Bad loans can significantly affect the financial stability and profitability of these institutions.
- By leveraging machine learning models, this project leverages financial dat a and predictive analytics to enhance decision making and optimize profita bility

PROBLEM DESCRIPTION

Current Challenges in Loan approval systems

Challenge	Why It's a Problem	What We Can Do
Slow decisions	Loan approvals take days due to ma nual checks	Use a model to give instant decision s at application time
Inconsistent approvals	Different staff may approve or rejec t the same applicant	Apply clear, data-based rules that tr eat everyone fairly
High risk of loan defaults	Approving risky applicants leads to financial losses	Predict approval more accurately to reduce bad loans

- *How can we make faster and more accurate approval decisions without increasing credit risk?*

INTERIM PRESENTATION SUMMARY

EDA :

✓ Approval more influenced by loan characteristics than demographics

✓ Zero approvals for previous defaulters

✓ Unexpected trends: higher interest rate or loan burden → **more approvals**

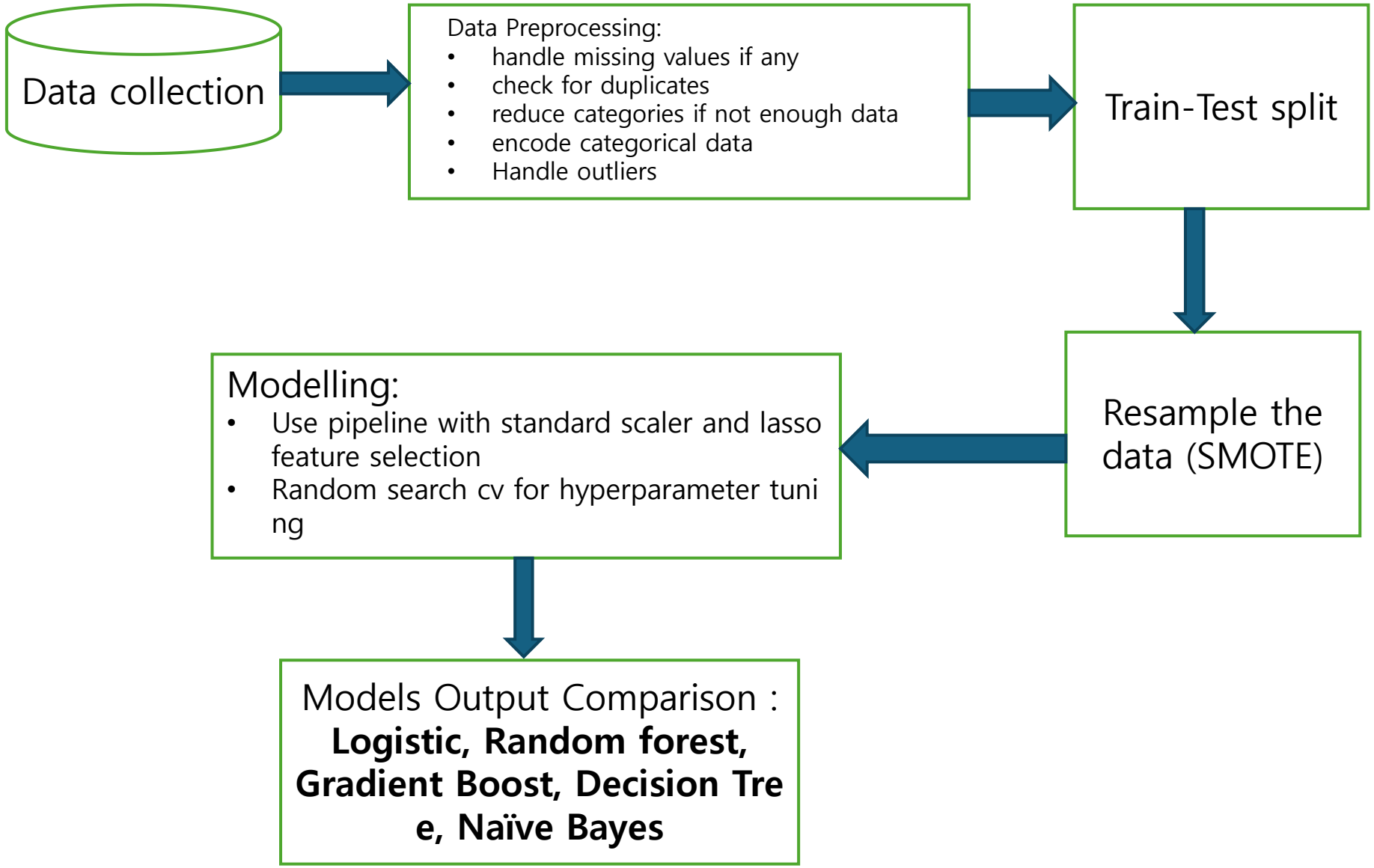
Monotonic & Non-linear patterns:

✓ Confirmed with Spearman & PDP plots

Modeling Direction:

✓ Developing interpretable models

MODEL PIPELINE

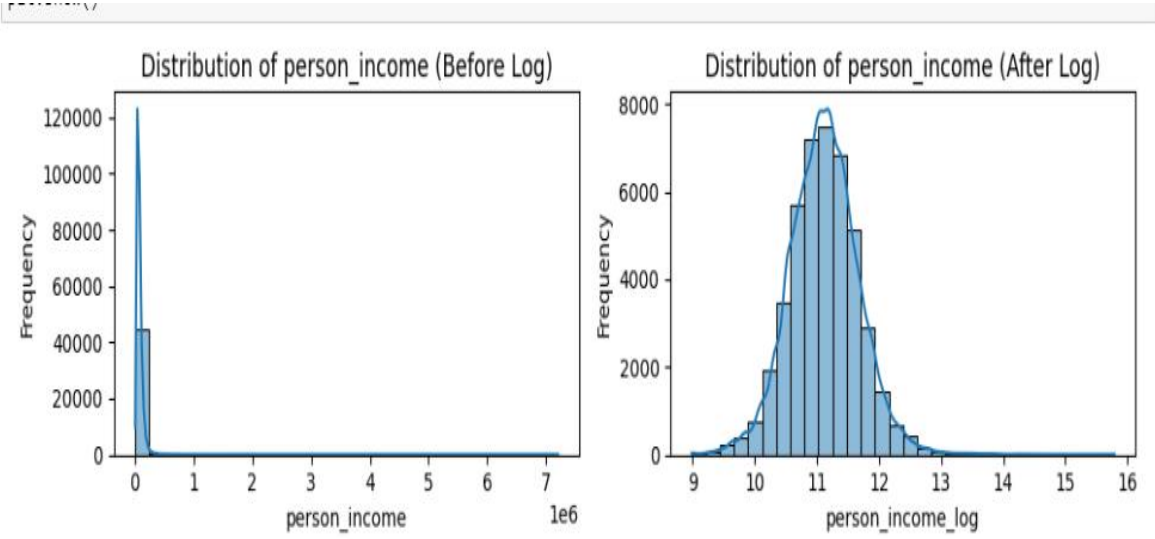


PRE-PROCESSING STEPS

1. Numerical Processing
   - ‘person_emp_exp’ - values greater than 55 were replaced with the medi an (uncommon for individuals to have over 55 years of work experience)
   - ‘Person_income’ – the columns is high skewed thus the values were tran sformed using log to normalize the distribution.

```
[20]: #check person experience
      data1['person_emp_exp'].describe()

Out[20]: count    45000.000000
         mean         5.410333
         std          6.063532
         min          0.000000
         25%          1.000000
         50%          4.000000
         75%          8.000000
         max        125.000000
         Name: person_emp_exp, dtype: float64
```



PRE-PROCESSING STEPS

2. Categorical Processing

- ‘person_education’ - combine ‘Masters’ & ‘doctorate’ and rename to’higher education. (*Doctorate value has less than 1% data thus considered not to have enough data to model on its own*)
- Categorical columns with more than 2 categories : one hot encode, else use dummy variable

Gender	Home_ownership
female	Rent
male	Mortgage
female	rent

Gender	Home_owenership_Rent	Home_owenership_Mortgage
0	1	0
1	0	1
0	1	0

PRE-PROCESSING STEPS

3. Other

- ‘Age’ - Encode

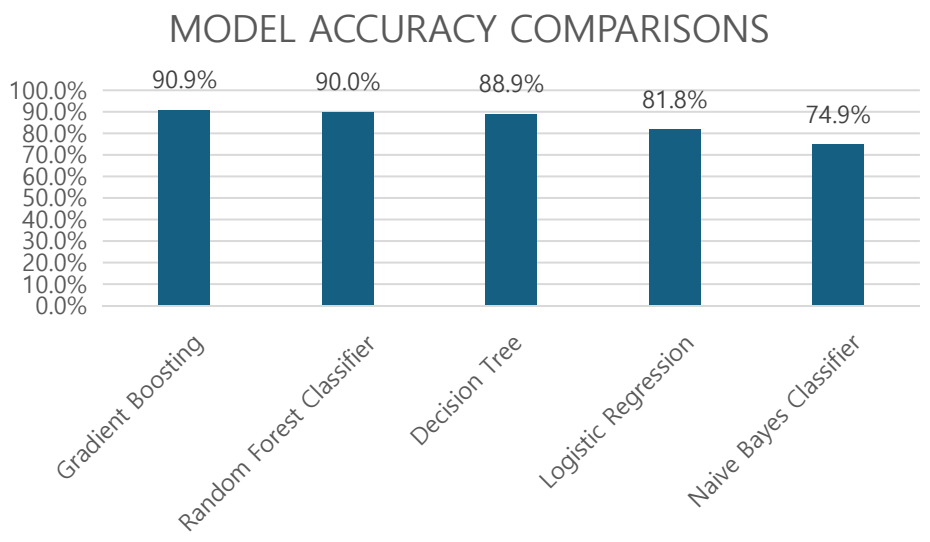
```
In [18]: data1["age_group"].value_counts(normalize=True)

Out[18]: age_group
         18-30    0.764422
         31+      0.235578
         Name: proportion, dtype: float64
```

- ‘previous_loan_defaults_on_file’ – Drop from data

The column is highly correlated to the target columns

MODEL PERFORMANCE – SUMMARY



MODEL	PRECISION	RECALL	F1_SCORE	ACCURACY
Gradient Boosting	**85.9%**	**70.5%**	**77.5%**	**90.9%**
Random Forest Classifier	83.6%	68.5%	75.3%	90.0%
Decision Tree	78.7%	68.9%	73.5%	88.9%
Logistic Regression	57.5%	70.5%	63.3%	81.8%
Naive Bayes Classifier	45.5%	63.2%	52.9%	74.9%

Gradient Boosting achieved:

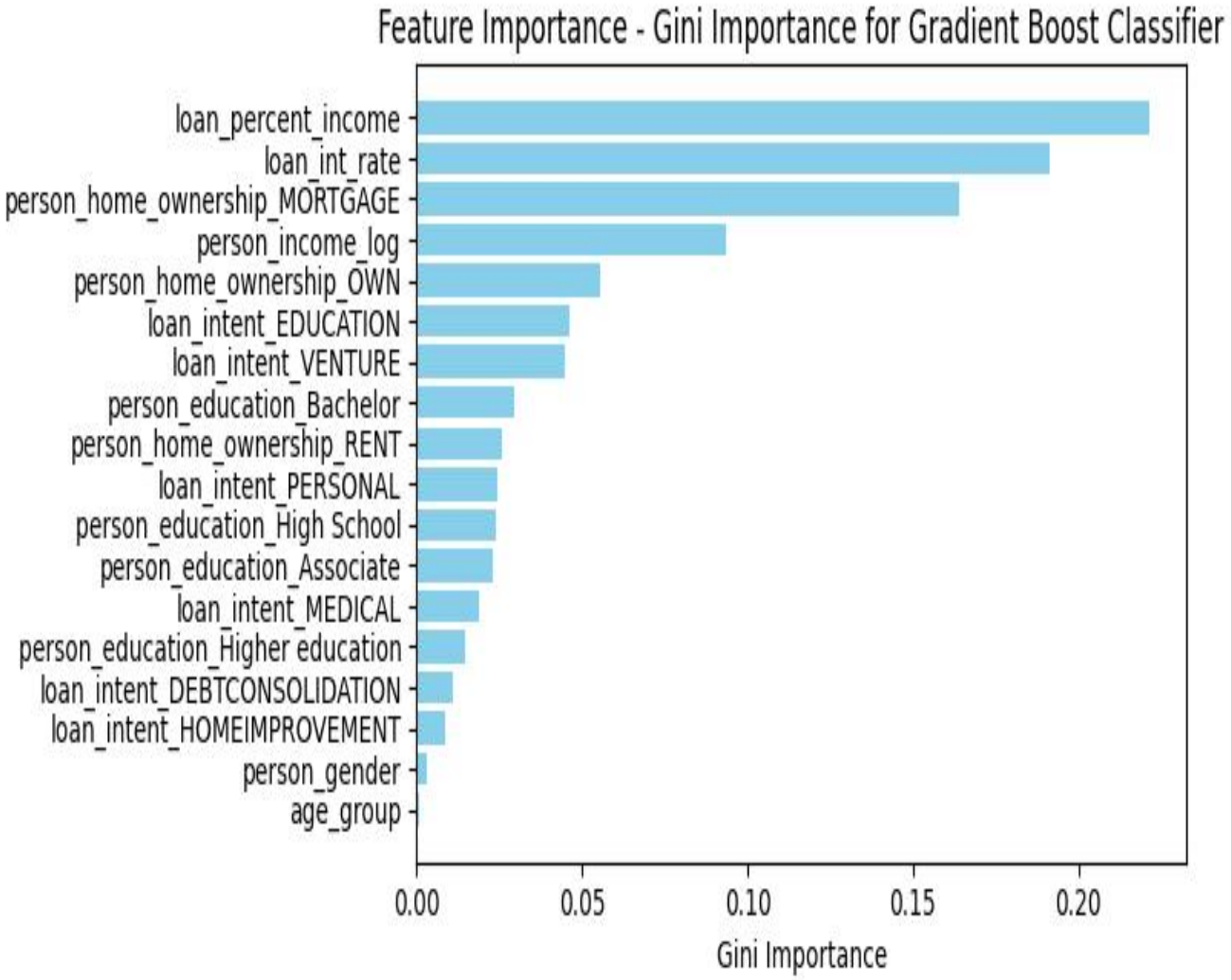
- Highest accuracy (**90.9%**)
- Best F1 score (**77.5%**)
- Strong precision (**85.9%**)

Random Forest was a close second in accuracy (**90.0%**) followed by Decision Tree Classifier

Logistic Regression had good recall (**70.5%**) but low precision (**57.5%**), indicating more false positives

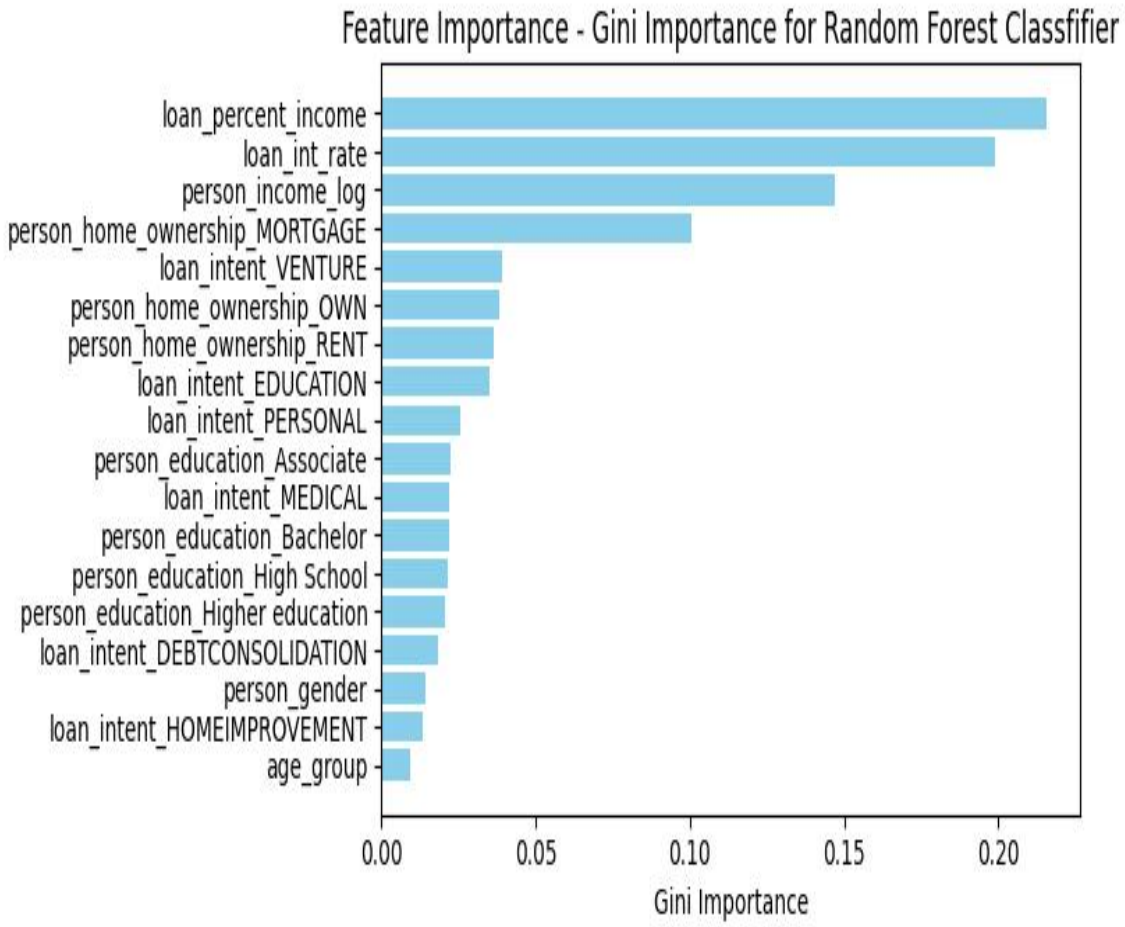
Naive Bayes underperformed across all metrics (**accuracy: 74.9%**)

MODEL INTEPRETATION– GRADIENT BOOST



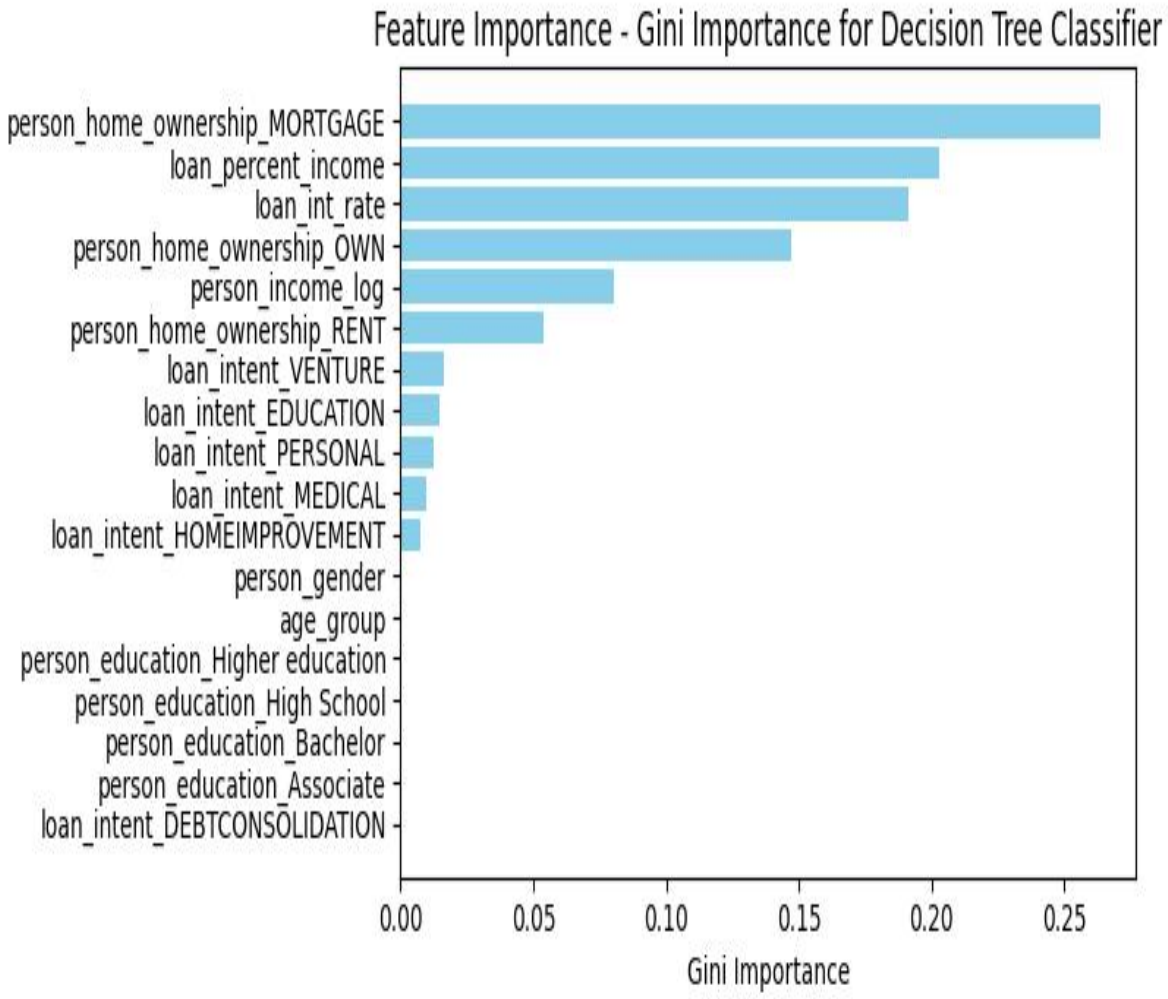
- ✓ *loan_percent_income and loan_int_rate* are the most influential features. Generally, high er interest or income burden signals risk.
- ✓ model captures **financial stability** through housing status.
- ✓ **intended use of funds** affects model risk perception
- ✓ As seen in the previous models, *person_edu cation_*, gender, and age_group* have **mini mal importance**.

MODEL INTEPRETATION – RANDOM FOREST



- ✓**loan_percent_income** is the most influential variable — reflects borrower's repayment burden
- ✓Purpose-related features (e.g., VENTURE, EDUCATION) carry lower but meaningful weight
- ✓**Education level** shows **minimal importance** — suggesting it plays a small role in RF decisions
- ✓**Demographics have very low influence**

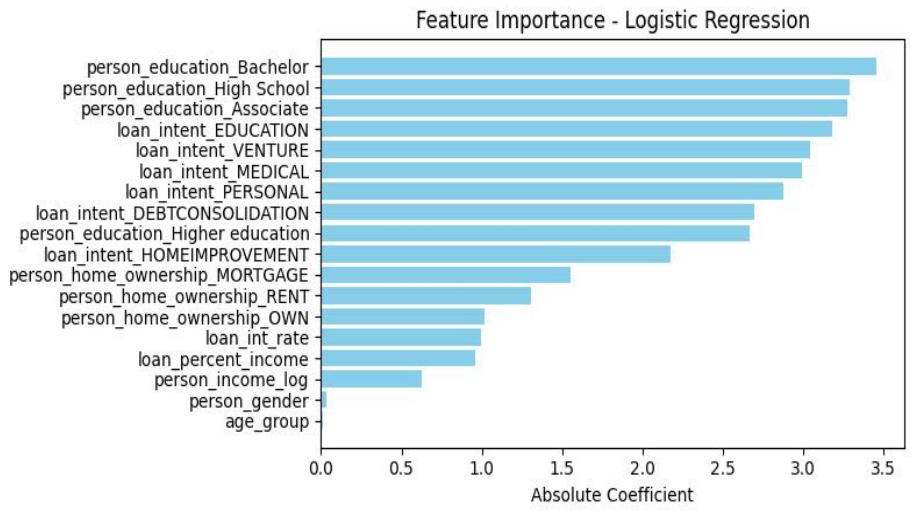
MODEL INTEPRETATION – DECISION TREE



- ✓ The tree heavily uses homeownership status as a pri mary split—interpretable as a proxy for financial sta bility or collateral.
- ✓ In contrast to logistic regression (where l*oan_perce nt_income and int_rate* had low coefficients), the tree finds them much more useful in splitting decisions.
- ✓ Unlike logistic regression, All *loan_intent_* and pe rson_education_** features show very low Gini importance.
- ✓ person_gender and age_group have very low importance

MODEL INTEPRETATION – LOGISTIC REGRESSION

	Feature	Coefficient (Abs)
3	person_education_Bachelor	3.457632
4	person_education_High School	3.289310
2	person_education_Associate	3.279113
10	loan_intent_EDUCATION	3.184372
14	loan_intent_VENTURE	3.041590
12	loan_intent_MEDICAL	2.991621
13	loan_intent_PERSONAL	2.876045
9	loan_intent_DEBTCONSOLIDATION	2.694793
5	person_education_Higher education	2.671293
11	loan_intent_HOMEIMPROVEMENT	2.174312
6	person_home_ownership_MORTGAGE	1.555028
8	person_home_ownership_RENT	1.305930
7	person_home_ownership_OWN	1.018379
16	loan_int_rate	0.998349
17	loan_percent_income	0.957622
15	person_income_log	0.625107
0	person_gender	0.030482
1	age_group	0.008337



✓ **Loan Purpose & Ownership**

Lenders routinely adjust pricing/approval by purpose (e.g., debt consolidation vs. venture).

Home-ownership signals collateral/security—moderate effect as e xpected.

❖ **Education's Outsized Influence**

Industry scorecards rarely weight education this heavily (regulatory/ fairness concerns).

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❖ **Underweighted Financial Ratios**

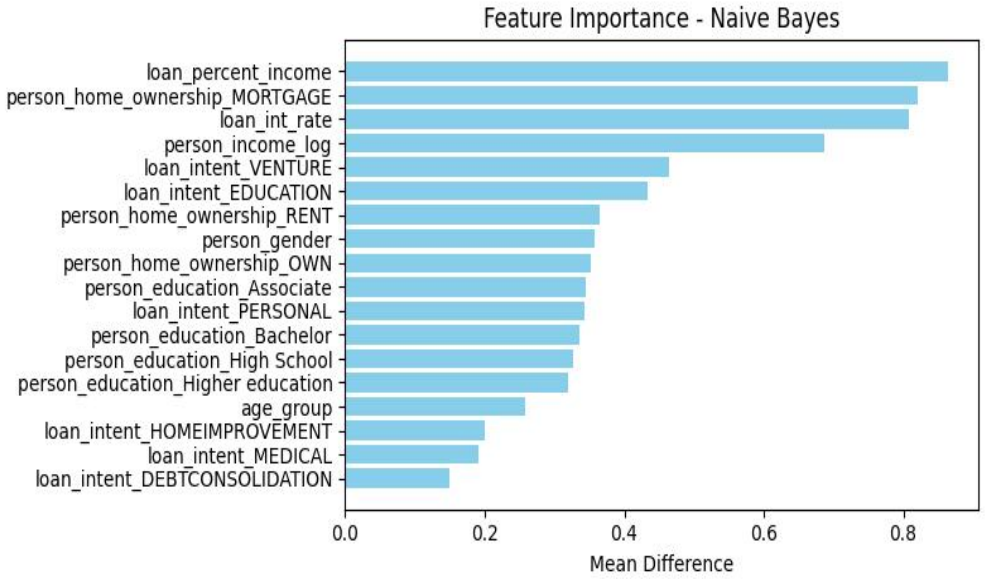
Debt-to-income and interest rate are core risk metrics, yet show only $|\beta| \approx 1.0$.

✓ **Demographics Minimal**

Gender & age nearly zero—great from a fairness- compliance standpoint

MODEL INTEPRETATION– NAIVE BAYES

	Feature	Mean Diff (Proxy Importance)
17	loan_percent_income	0.864312
6	person_home_ownership_MORTGAGE	0.820424
16	loan_int_rate	0.807520
15	person_income_log	0.686880
14	loan_intent_VENTURE	0.464021
10	loan_intent_EDUCATION	0.433695
8	person_home_ownership_RENT	0.364932
0	person_gender	0.357495
7	person_home_ownership_OWN	0.352819
2	person_education_Associate	0.344824
13	loan_intent_PERSONAL	0.343360
3	person_education_Bachelor	0.335053
4	person_education_High School	0.326927
5	person_education_Higher education	0.320200
1	age_group	0.258085
11	loan_intent_HOMEIMPROVEMENT	0.200762
12	loan_intent_MEDICAL	0.190807
9	loan_intent_DEBTCONSOLIDATION	0.149805



- ✓ *loan_percent_income, loan_int_rate, and person_income_log* dominate the ranking showing high class separation.
- ✓ *person_home_ownership_MORTGAGE* is the 2nd most influential. Suggests MORTGAGE status differentiates approved vs rejected clearly—possibly due to assumed creditworthiness.

COMPARATIVE MODEL INTERPRETATION

Feature	EDA Insight / Domain Expectation	Logistic Regression	Decision Tree	Gradient Boost	Naive Bayes
loan_percent_income	▲Higher for approved loans; key affordability ratio	✕ Low	✓ High	✓ Highest	✓ Highest
loan_int_rate	▲Higher in approved loans (counterintuitive pricing signal)	✕ Low	✓ High	✓ High	✓ High
person_income_log	▲Higher in approved cases	✕ Low	⚠ Moderate	✓ Moderate	✓ High
person_home_ownership_MORTGAGE	▲Seen as financially stable, collateral-backed	✓ Moderate	✓ Top Feature	✓ Top 3	✓ High
person_home_ownership_RENT/OWN	Mixed effect, varies by model	✓ Moderate	✓ Moderate	✓ Moderate	✓ Moderate
loan_intent_EDUCATION	⛔ Riskier, lower approval in EDA	✓ Very High	✕ Low	⚠ Mid Importance	✓ High
loan_intent_VENTURE	⛔ Riskier, lowest approval	✓ Very High	✕ Low	⚠ Mid Importance	✓ High
loan_intent_DEBTCONSOLIDATION	✓ Safest, highest approval in EDA	✓ Moderate	✕ Lowest	✕ Lowest	✕ Very Low
person_education_*	⛔ Should have minimal impact, fairness issue	✕ Too High	✓ Low	✓ Low	✓ Low
person_gender, age_group	⚠ Fairness concern, should be minimal	✓ Very Low	✓ Very Low	✓ Very Low	✓ Very Low

MODELS' RESULTS SUMMARY

- ✓ **Model interpretations reinforce EDA findings and real-world credit logic**. Features like *loan_percent_income, loan_int_rate, and person_income_log* consistently emerged as top predictors, aligning with how lenders assess affordability and risk.
- ✓ Tree-based models also captured the **impact of loan purpose and home ownership** — features linked to collateral and loan intent in financial practice.
- ✓ **This coherence between EDA, domain expectations, and model outputs supports the reliability and interpretability of the selected models**.

KEY MISSED FEATURE

CREDIT SCORE

- ✓ While it exists in the dataset and should be central, it's consistently low-ranked across all models.

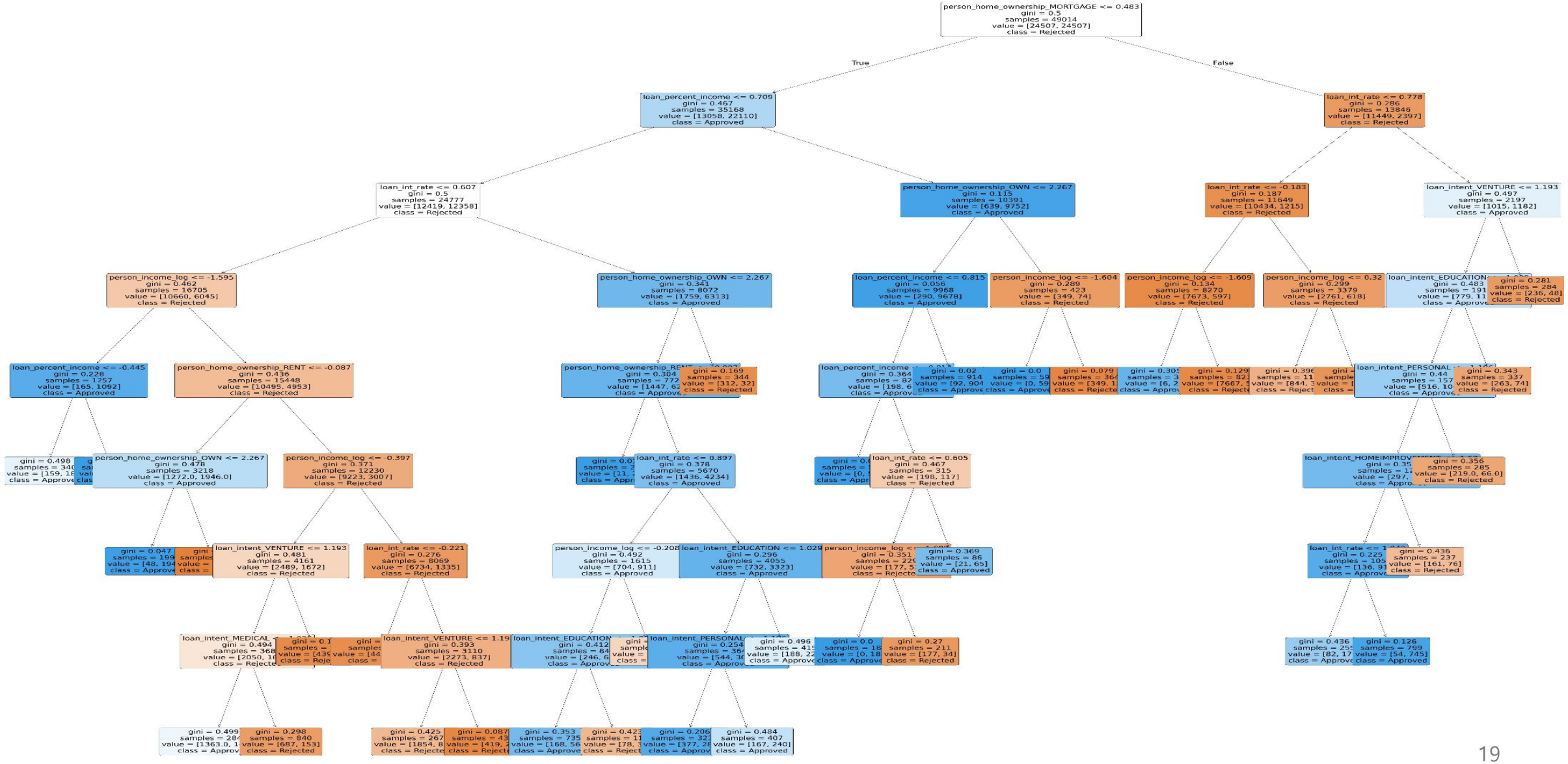
Possible Reasons:

- ✓ Redundancy with *loan_int_rate* and *loan_percent_income* (credit score may already affect interest rate assigned).
- ✓ No nonlinearity captured (e.g., good scores above 700 might all behave the same).
- ✓ Discretization or skew issues may flatten its influence.

COLOR DECISION TREE

- Color DT visualizes decision rules using color-coded leaf nodes based on predicted class and confidence
- Clear thresholds and paths make model logic **human-readable**
- Derived **compact rules** simplify the full paths by removing redundant splits while preserving prediction fidelity

COLOR DECISION TREE



COLOR DECISION TREE CONDITIONS

CONDITIONS	PREDICTION
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate <= 0.61 & person_income_log <= -1.60 & loan_percent_income <= -0.45	Approved
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate <= 0.61 & person_income_log <= -1.60 & loan_percent_income > -0.45	Approved
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate <= 0.61 & person_income_log > -1.60 & person_home_ownership_RENT <= -0.09 & person_home_ownership_OWN <= 2.27	Approved
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate <= 0.61 & person_income_log > -1.60 & person_home_ownership_RENT <= -0.09 & person_home_ownership_OWN > 2.27	Rejected
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate > 0.61 & person_home_ownership_OWN <= 2.27 & person_home_ownership_RENT <= -0.09	Approved
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income <= 0.71 & loan_int_rate > 0.61 & person_home_ownership_OWN > 2.27	Rejected
person_home_ownership_MORTGAGE <= 0.48 & loan_percent_income > 0.71 & person_home_ownership_OWN <= 2.27 & loan_percent_income <= 0.81 & loan_percent_income <= 0.81	Approved
person_home_ownership_MORTGAGE <= 0. 48 & loan_percent_income > 0.71 & person_home_ownership_OWN <= 2.27 & loan_percent_income <= 0.81 & loan_percent_income > 0.81 & loan_int_rate > 0.61	Approved

- ✓ First split on person_home_ownership_MORTGAGE
  → Indicates mortgage status is a strong first-level discriminator
- ✓ Lower loan_percent_income and loan_int_rate consistently lead to approval paths
- ✓ High person_income_log and education-related intent (loan_intent_EDUCATION) contribute to approval
- ✓ Reject paths often triggered by:
  - •High loan burden or rate
  - •Low income
  - •Riskier loan intents (e.g., VENTURE, PERSONAL)

COLOR DECISION TREE – COMPACT RULES

```
Rule 1
Original Rule:
IF person_home_ownership_MORTGAGE <= 0.5 AND loan_percent_income <= 0.23 AND loan_i
nt_rate <= 13.49 AND person_income_log <= 10.121 AND loan_percent_income <= 0.12 TH
EN class = Approved (RF: 1.0)

Compact Rule (Method 1):
IF person_home_ownership_MORTGAGE <= 0.5 AND loan_percent_income <= 0.12 AND loan_i
nt_rate <= 13.49 AND person_income_log <= 10.121 THEN class = Approved (RF: 1.0)

Rule 4
Original Rule:
IF person_home_ownership_MORTGAGE <= 0.5 AND loan_percent_income <= 0.23 AND loan_i
nt_rate <= 13.49 AND person_income_log > 10.121 AND person_home_ownership_RENT <= 0
.5 AND person_home_ownership_OWN > 0.5 THEN class = Rejected (RF: 0.0)

Compact Rule (Method 1):
IF person_home_ownership_MORTGAGE <= 0.5 AND loan_percent_income <= 0.23 AND loan_i
nt_rate <= 13.49 AND person_income_log > 10.121 AND person_home_ownership_RENT <= 0
.5 AND person_home_ownership_OWN > 0.5 THEN class = Rejected (RF: 0.0)
```

RULE 1: High Confidence Approval Rule (RF: 1.0)

- A clean, low-debt, moderate-income applicant with no mortgage is always approved.
- Compact simplification removed a redundant loan_percent_income <= 0.23 condition without changing the outcome.

RULE 4: Consistently Wrong Rejection Rule (RF: 0.0)

- The model incorrectly predicts approval, leading to **0% rule fidelity**
- Likely a contradictory input (e.g., OWN & RENT both 1) or outlier. The model learned to approve these cases despite the rule logic.

DT VS COLOR DT AVERAGE ACCURACY

MODEL	Accuracy
DECISION TREE	88.9%
WEIGHTED DT COLOR	85.1%

- **Weighted accuracy** is used for Color DT because each rule covers a different number of samples

  → This gives a fairer overall measure based on how impactful each rule is

- Color DT achieves **85.1% weighted accuracy**, close to the full Decision Tree's 88.9%
- Despite simplifying paths, the **compact rules retain key decision logic**
- Slight accuracy trade-off is acceptable for the gain in **interpretability and transparency**

FINAL SUMMARY & CONCLUSION

Project Goal

✓ Boost loan-approval **accuracy** & **transparency** in high-risk finance via ML + EDA.

Top Model

✓ **Gradient Boosting** — 90.9 % accuracy, 77.5 % F1 (Random Forest close: 90.0 %).

Key Predictors

✓ loan_percent_income, loan_int_rate, person_income_log, Mortgage status.

✓ Demographics minimal → fairness-compliant.

Interpretability Wins

✓ Color DT + compact rules keep **85.1 % weighted accuracy** (vs 88.9 % full tree) while greatly improving readability.

Business Impact

✓ Faster, consistent approvals; lower default risk; audit-ready rationale.

Future Considerations

✓ enrich extreme credit-score data, test fairness-enhanced boosting.

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